The importance of anomaly in neurolinguistic research

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Abstract. Anomaly has always attracted the interest of language scholars, either as a possible mechanism for language change or as a window into speakers' judgments and competence. Here I will discuss the use of errors in neurolinguistics, focusing on Moro's prolific ideas to disentangle grammar in the brain by presenting selective anomalies to neurotypical individuals as well as clinical groups. This prompts further considerations of the nature of anomaly for the study of language as a whole, from its core structures to its most creative nuances.

> Somehow it seems to fill my head with ideas–only I don't exactly know what they are!
>
> Alice's reaction to the poem "Jabberwocky", Lewis Carroll, *Through the Looking-Glass, and what Alice Found There*

"Il gulco gianigeva le brale" (*The gulk ganfeld the brals*) is a well-known phrase for those interested in neurolinguistics. It was used as a control condition in a study by Moro and colleagues (Moro et al. 2001), where Italian sentences made up of pseudowords (i.e., strings of letters that resemble real words but are presumably meaningless, as traditionally employed in psycholinguistics; Taft & Forster 1975) and displaying different types of error (phonological, as in "Il gulco gianigzleva…", with the illegal strings of consonant "gzl"; morpho-syntactic, as in "Il gulco ha gianigiata…", with the "-a$_{\mathrm{F.SG}}$" marking on the past participle instead of "-o$_{\mathrm{M.SING}}$"; or syntactic, as in "Gulco il gianigeva le brale", with the wrong noun-determiner order) were presented to Italian-speaking participants while recording their brain activity via functional magnetic resonance imaging (fMRI). Results highlighted a specific pattern of brain activations when participants were processing syntactic errors, involving Broca's area. The study had a great influence in claiming the syntactic specialization of Broca's area, extending previous literature from the same period that had also used errors (see, for instance, grammatical vs. spelling errors in Embick et al. 2000). The study of Moro and colleagues widened the typology of errors and combined them with the use of pseudowords, thus preventing any confound from the semantic level, besides making the *gulk* sentence so iconic.

What's the rationale behind the use of errors in neurolinguistic research? One of the biggest aims of neurolinguistics is to test whether different levels of linguistic representation are grounded in the brain. In a shift from the classic aphasiological model that assumed different neural correlates for different modalities of language use (e.g., production vs. comprehension), modern neurolinguistics has from its start aimed at finding the extent to which levels of processing (e.g., syntax vs. semantics), as well as linguistic categories (e.g., nouns vs. verbs), correspond to different brain responses (Bambini 2012; Bambini & Canal 2021). Given this aim, it is necessary to elicit the functioning of the specific network supposedly devoted to processing the linguistic level under consideration. This is especially challenging, because in language the different layers of representation are given all at once: syntactic structures are entangled with morphological markers, and in turn, grammatical constructions are entangled with meaning. To solve this issue, a common strategy is to employ errors that selectively violate one linguistic component. For instance, early studies on the electrophysiological response to language reshaped the classic oddball paradigm and employed errors consisting in a semantic deviation (e.g., the famous *He spread the warm bread with socks*; Kutas & Hillyard 1980) or a syntactic violation (e.g., *The woman persuaded to…* as compared to *The woman struggled to…*; Osterhout & Holcomb 1992). The assumption behind the use of errors is that they would be detected by engaging the network responsible for processing the rules and constructions belonging to the level of linguistic representation that is violated (Moro 2004). Note that these studies often include a baseline condition (e.g., detecting irregularities in a string of symbols) to factor out more basic error detection mechanisms and ensure that the observed response reflects specific linguistic processes. Note also that linguistic errors were shown not to be necessary or sufficient to elicit a certain response, as illustrated by N400 effects for semantic manipulations not dealing with anomalies (Kutas & Federmeier 2011), but to tax the processing system in a quantitatively greater fashion than well-formed stimuli (for a critical perspective, see Hasson et al. 2018). A slightly different perspective on errors is taken by theoretical research in linguistics, where sentences containing errors have been used to corroborate linguists' intuitions and test the speakers' competence. Yet weak quantitative standards were pointed out for this approach (Gibson & Fedorenko 2010), an issue that neurolinguistic studies have mostly overcome.

Throughout his studies, Moro blended these research traditions and extended the theoretical capacity of the error paradigm to test not simply the brain specialization for syntax but also the limits of human syntax and its apparatus of rules. In another study from 2003 (Musso et al. 2003), errors violating key principles of Universal Grammar were presented to participants in the context of a language learning task. To give a bit more details, participants were German native speakers with no prior knowledge of Italian or Japanese: before the scan, they were taught a series of words in these foreign languages, and then, during the scan, they read sentences composed of these words, which either followed Universal Grammar (e.g., containing a null subject, as in "Mangio la pera", *(I) eat the pear*) or violated it, being based on mere linear order (e.g., containing a negation placed always after the third word in the sentence, as in "Paolo mangia la no pera", *Paolo eats the no pear*). Results highlighted that only when participants were learning languages with rules that were compatible with the Universal Grammar, Broca's area activated, while languages with errors not following Universal Grammar, that is, *impossible* languages, engaged a different apparatus (involving the right hemisphere), related to retrieving information from episodic memory. Via errors, it was shown that the human brain is fine-tuned to recognize only languages with a specific set of rules.

Because of its capacity to draw the borders of our faculty of language, the error paradigm can offer also a privileged window into a number of brain mechanisms associated with linguistic processing. Among these, we can include learning mechanisms, as the experiment on artificial languages described above illustrates. Referring now to naturally occurring learning, the literature showed that, for instance, L2 speakers are less sensitive to grammatical violations, as reflected in a reduced amplitude of the P600 components compared to native speakers (Rossi et al. 2006). Yet this pattern changes depending on individual proficiency, as the higher the competence in the second language the greater the sensitivity to the errors. But what if the errors tap into the constituent structure and are combined with a shift between languages? Moro contributed to a study on French-Italian bilinguals (Abutalebi et al. 2007) where participants were exposed to switches that could either respect the constituent structure (e.g., occurring between the noun and the verb, as in "Il piccolo principe_est allé", *The little prince_was going*; language switch marked with the underscore) or break it (e.g., occurring between the auxiliary verb and the participle of the lexical verb, as in "J'ai_risposto", *I have_answered*). Only switches that break the constituent structure generated the pattern of brain activations typically observed for syntax, including Broca's area, whereas switches respecting the constituent structure were associated with lexico/semantic processes. This finding not only highlights the neural correlates of switching phenomena, but also explains why switches tend to occur in certain positions rather than others, particularly at major syntactic and prosodic boundaries (Schmid 2005), where they are processed as lexical alternatives.

Furthermore, errors can shed light on the disruption of the language system in neurological and psychiatric conditions. In the broad field of language disorder, from classic studies on aphasia to more recent investigations in other clinical groups, great attention has been devoted to analyzing errors. We know, for instance, that English-speaking aphasic patients tend to produce grammatical errors such as *boy* for *boys* or *run* for *runs*, omitting the bound morphemes, whereas Italian-speaking ones might produce "bambino correre" (*child run*), selecting the incorrect form of the inflectional paradigm (Bambini & Canal 2021). As evident from these examples, the interest is in the analysis of the errors produced by the individuals in spontaneous or elicited speech. But what if we use errors to test the vulnerability of linguistic competence via anomaly judgment task? This can be particularly helpful in clinical groups that do not exhibit marked aphasic symptoms in spontaneous production but might have impaired linguistic competence, such as in mental illness and some degenerative brain disorders. Moro and colleagues used this approach – not common in the literature (but see (Grodzinsky & Finkel 1998; S. M. Wilson & Saygın 2004) – in a sample of individuals with schizophrenia, who were asked to judge the acceptability of sentences containing different syntactic errors. Notably, errors violated Universal Grammar principles (e.g., involving violations of locality principles, as in the question "Chi Giovanni vuole contattare l'infermiera prima di incontrare?", *Who does John want to contact the nurse before meeting?*) and were not intertwined with morphological operations (Moro et al. 2015). Results highlighted a significantly lower accuracy in detecting syntactic errors in the clinical group compared to the matched control group, not related to working memory. Similar findings were obtained in another study testing the syntactic competence in fronto-temporal dementia spectrum, where patients with corticobasal degeneration syndrome were found to be impaired in syntactic knowledge, while exhibiting intact sentence comprehension (Cotelli et al. 2007). Collectively, these studies point to the vulnerability of syntactic competence, even when other aspects of language (e.g., lexico/semantic knowledge) might support sentence comprehension and production, thus offering evidence in favor of the independence of such syntactic competence. Interestingly, the Moro et al. 2015 study also employed a novel type of semantic anomaly, not based on semantic clashes, but involving a contradiction within the same semantic domain (e.g., "Ho asciugato la maglietta con l'acqua", *I have dried my new shirt with water*). The detection of these latter anomalies was preserved in schizophrenia, which points to an interesting dissociation between syntax and semantics, and leave room for attributing interpretative difficulties in schizophrenia to the breakdown of higher levels of processing such as pragmatics (Bambini et al. 2016).

The paragraphs above illustrate the power of errors for neurolinguistics research. I see the use of the error paradigm in neurolinguistics as the last step of a long history of language studies that reflected on deviant stimuli, from the quarrel of ancient grammarians between anomaly and analogy to studies on sociolinguistic variation that identify errors as a motor of language change (Labov 1963; for a neurolinguistic perspective, see Bambini et al. 2021), up to jocular uses of errors in children's literature such as Lewis Carroll's poem *Jabberwocky* (Carrol 1871) or the nursery rhymes by Gianni Rodari (Rodari 1964; see also De Santis 2021). Andrea Moro exploited this history of thought in its full capacity to explore the limits of human linguistic competence, or, as he would say, the boundaries of Babel. An open question remains, however, whether errors can be generated for all layers of language. While native speakers of English would judge *John eats no apple* as agrammatical, how about a sentence such as *John buried the guitar*? Several neurolinguistic studies have compared different anomalies from the phonological up to the pragmatic level (Kuperberg, McGuire & David 2000). Pragmatic anomalies used in the literature are things such as the "guitar" example above, which violates word knowledge by combining incompatible semantic fields. Yet we can easily imagine a context where this sentence becomes meaningful and pragmatically appropriate, where John metaphorically buried the guitar, meaning that he quit playing it. We might be facing an interesting division in our language competence. While the syntactic component seems to be rather stable, our pragmatic competence might be more malleable and tolerant. In other words, while Universal Grammar has specific boundaries, interpretative mechanisms seem to be fine-tuned to overcome these boundaries, seek relevance and accommodate utterances in communication (Wilson & Sperber 2006). In this view, even *Colorless green ideas sleep furiously* might become acceptable, because either we interpret it metaphorically (Bambini 2017) or we overlook it in conversation (Galantucci, Roberts & Langstein 2018). In a very recent article, Greco et al. 2023 introduced the notion of gradable predictability, distinguishing between stimuli that are immediately predictable and those that are less predictable due to grammatical constraints. Predictability (or its mirror concept of surprisal) is largely intertwined with anomaly, since errors are normally unexpected (indeed, they typically represent a limited proportion of stimuli in experimental paradigms). The gradable distinction might thus be used for describing variation across linguistic anomalies more broadly, extending to the pragmatic and interpretative domain, to capture the interplay of competence and prediction mechanisms in language. This appears as a promising avenue to further characterize the divide, as well as the resemblance, between syntax and pragmatics and the architecture of language in general.

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